

# Michael Gruenstaeudl (Grünstäudl), PhD

📍 Fort Hays State University, 600 Park Street, Hays, KS 67601, USA

✉ [m\\_gruenstaeudl@fhsu.edu](mailto:m_gruenstaeudl@fhsu.edu) | 🖱 [gruenstaeudl-lab.com](http://gruenstaeudl-lab.com) | 🆔 0000-0002-1666-1773

## Education and Professional Positions

### Education

|                                                                                                                            |                                    |      |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|
| <b>Habilitation</b> in Bioinformatics and Botany                                                                           | Freie Universität Berlin, Germany  | 2023 |
| Habilitation thesis: "Development and application of bioinformatic tools toward process automation in plant phylogenetics" |                                    |      |
| <b>Ph.D.</b> in Plant Biology                                                                                              | University of Texas at Austin, USA | 2013 |
| <b>M.Sc.</b> in Plant Biology                                                                                              | University of Vienna, Austria      | 2007 |

### Professional Positions

|                                                                                 |                                   |              |
|---------------------------------------------------------------------------------|-----------------------------------|--------------|
| <b>Assistant Professor (Tenure-Track)</b><br>Dept. Biological Sciences          | Fort Hays State University, USA   | 2023–ongoing |
| <b>Postdoctoral Researcher</b><br>Dept. Biology, Chemistry, Pharmacy            | Freie Universität Berlin, Germany | 2015–2022    |
| <b>Postdoctoral Researcher</b><br>Dept. Evolution, Ecology & Organismal Biology | Ohio State University, USA        | 2014–2015    |

## Research

### Research Interests

Phylogenomics and bioinformatics of evolutionary genomic data • Computational analysis of large-scale sequence datasets • Systematics and evolutionary biology of plants, algae, and cyanobacteria

### Research Funding / Grants

- [8] M Gruenstaeudl. *Genome sequencing of bloom-forming cyanobacteria in western Kansas*. Data science core / PUI grant. Single PI; Grant no. P20GM103418/GR00848; DURATION: 2025–ongoing; AMOUNT: **\$ 33,210**. FUNDER: Kansas INBRE, National Institutes of Health, USA, 2025.
- [7] M Gruenstaeudl. *Implementing a course-based undergraduate research experience to train students in identifying bloom-forming cyanobacteria using state-of-the-art genetic techniques*. Research grant. Single PI; Grant no. GR00807; DURATION: 2025–ongoing; AMOUNT: **\$ 24,059**. FUNDER: Kansas Water Office, Fort Hays State University, USA, 2025.
- [6] M Gruenstaeudl. *Enabling data quality assessment of organelle genomes archived on GenBank through novel open-source software tools*. R01 research grant. Single PI; Grant no. R01LM014506; DURATION: 2024–ongoing; AMOUNT: **\$ 239,206**. FUNDER: NLM, National Institutes of Health, USA, 2024. URL: [https://reporter.nih.gov/search/SLep2Yj6r0WWAVF8ECi\\_HA/project-details/10858197](https://reporter.nih.gov/search/SLep2Yj6r0WWAVF8ECi_HA/project-details/10858197).

- [5] TL Phelps-Durr, M Gruenstaeudl, LE Patrick, SD Elzay, and NK Lazerus. *Uncovering change catalysts: Examining Community of Practice members as they leverage individual course improvements to achieve systemic departmental change*. IUSE: EDU research grant. Co-PI; Grant no. 2417083; DURATION: 2024–ongoing; AMOUNT: **\$ 385,971**. FUNDER: National Science Foundation, USA, 2024. URL: [https://www.nsf.gov/awardsearch/show-award/?AWD\\_ID=2417083](https://www.nsf.gov/awardsearch/show-award/?AWD_ID=2417083).
- [4] M Gruenstaeudl. *Data mining of organellar genomes using bioinformatics—A powerful method to understand the quality of publicly accessible organellar sequence records*. Start-up grant. Single PI; Grant no. P20GM103418/GR509061; DURATION: 2023–2024; AMOUNT: **\$ 24,765**. FUNDER: Kansas INBRE, National Institutes of Health, USA, 2023.
- [3] M Gruenstaeudl. *Plastid phylogenomics of water lilies, with a focus on partitioning strategies and the development of bioinformatic tools*. Research grant (Sachbeihilfe). Single PI; Grant no. 418670221; DURATION: 2018–2022; AMOUNT: **€ 69,360**. FUNDER: Deutsche Forschungsgemeinschaft, Germany, 2018. URL: <https://gepris.dfg.de/gepris/projekt/418670221?language=en>.
- [2] M Gruenstaeudl. *Plastid phylogenomics of water lilies (Nymphaeaceae) and early-diverging sunflowers (Barnadesioideae)*. Seed grant. Single PI; Grant no. 21224600; DURATION: 2016–2018; AMOUNT: **€ 11,470**. FUNDER: Forschungskommission der Freien Universität Berlin, Germany, 2016.
- [1] M Gruenstaeudl. *Graduate Research Fellowship*. Research fellowship. Single PI; Grant no. F816842; DURATION: 2011–2012; AMOUNT: **\$ 26,772**. FUNDER: University of Texas at Austin, USA, 2011.

## Publications

Graduate and undergraduate student mentees are underlined

- [27] N Jenke, Smith, GM, Magar Thapa, B, and **Gruenstaeudl, M**. “Variation of and associations with the depth and evenness of sequencing coverage in archived plastid genomes”. In: *Scientific Reports* 15 (2025). <https://doi.org/10.1038/s41598-025-11568-9>, p. 26294.
- [26] JA Roestel, JH Wiersema, RK Jansen, T Borsch, and **Gruenstaeudl, M**. “On the importance of sequence alignment inspections in plastid phylogenomics – an example from revisiting the relationships of the water-lilies”. In: *Cladistics* 40 (2024). <https://doi.org/10.1111/cla.12584>, pp. 469–495.
- [25] E Giorgashvili, K Reichel, C Caswara, V Kerimov, T Borsch, and **Gruenstaeudl, M**. “Software choice and sequencing coverage can impact plastid genome assembly – A case study in the narrow endemic *Calligonum bakuense*”. In: *Frontiers in Plant Science* 13 (2022). <https://doi.org/10.3389/fpls.2022.779830>, p. 779830.
- [24] B Escobari, T Borsch, TS Quedensley, and **Gruenstaeudl, M**. “Plastid phylogenomics of the Gynoxoid group (Senecioneae, Asteraceae) highlights the importance of motif-based sequence alignment amid low genetic distances”. In: *American Journal of Botany* 108 (2021). <https://doi.org/10.1002/ajb2.1775>, pp. 2235–2256.
- [23] Mehl, T and **Gruenstaeudl, M**. “airpg: Automatically accessing the inverted repeats of archived plastid genomes”. In: *BMC Bioinformatics* 22 (2021). <https://doi.org/10.1186/s12859-021-04309-y>, p. 413.
- [22] I Duran, A Marrero, F Msanda, C Harrouni, **Gruenstaeudl, M**, J Patino, J Caujape-Castells, and C Garcia-Verdugo. “Iconic, threatened, but largely unknown: Biogeography of the Macaronesian dragon trees (*Dracaena* spp.) as inferred from plastid DNA markers”. In: *Taxon* 69 (2020). doi: <https://doi.org/10.1002/tax.12215>, pp. 217–233.
- [21] **Gruenstaeudl, M**. “annonex2embl: automatic preparation of annotated DNA sequences for bulk submissions to ENA”. In: *Bioinformatics* 21 (2020). doi: <https://doi.org/10.1093/bioinformatics/btaa209>, p. 207.

- [20] **Gruenstaeudl, M** and **Jenke, N**. "PACVr: Plastome Assembly Coverage Visualization in R". In: *BMC Bioinformatics* 36 (2020). doi: <https://doi.org/10.1186/s12859-020-3475-0>, pp. 3841–3848.
- [19] **Gruenstaeudl, M**. "Why the monophyly of Nymphaeaceae currently remains indeterminate: An assessment based on gene-wise plastid phylogenomics". In: *Plant Systematics and Evolution* 305 (2019). doi: <https://doi.org/10.1007/s00606-019-01610-5>, pp. 827–836.
- [18] **Gruenstaeudl, M** and **Hartmaring, Y**. "EMBL2checklists: A Python package to facilitate the user-friendly submission of plant and fungal DNA barcoding sequences to ENA". In: *PLoS ONE* 14 (2019). doi: <https://doi.org/10.1371/journal.pone.0210347>, e0210347.
- [17] A Szukala, N Korotkova, **Gruenstaeudl, M**, AN Sennikov, GA Lazkov, SA Litvinskaya, SA Gabrielian, T Borsch, and E von Raab-Straube. "Phylogeny of the Eurasian genus *Jurinea* (Asteraceae: Cardueae): Support for a monophyletic genus concept and a first hypothesis on overall species relationships". In: *Taxon* 68 (2019). doi: <https://doi.org/10.1002/tax.12027>, pp. 112–131.
- [16] V Di Vincenzo, **Gruenstaeudl, M**, L Nauheimer, M Wondafrash, P Kamau, S Demissew, and T Borsch. "Evolutionary diversification of the African achyranthoid clade (Amaranthaceae) in the context of sterile flower evolution and epizoochory". In: *Annals of Botany* 122 (2018). doi: <https://doi.org/10.1093/aob/mcy055>, pp. 69–85.
- [15] **Gruenstaeudl, M**, N Gerschler, and T Borsch. "Bioinformatic workflows for generating complete plastid genome sequences - An example from *Cabomba* (Cabombaceae) in the context of the phylogenomic analysis of the water-lily clade". In: *Life* 8 (2018). doi: <https://doi.org/10.3390/life8030025>, p. 25.
- [14] TS Quedensley, **Gruenstaeudl, M**, and RK Jansen. "Phylogenetic relationships of the Mexican tussilaginoide genera (Asteraceae: Senecioneae)". In: *Journal of the Botanical Research Institute of Texas* 12 (2018), pp. 481–498. ISSN: 1934-5259.
- [13] **Gruenstaeudl, M**, BC Carstens, A Santos-Guerra, and RK Jansen. "Statistical hybrid detection and the inference of ancestral distribution areas in *Tolpis* (Asteraceae)". In: *Biological Journal of the Linnean Society* 121 (2017). doi: <https://doi.org/10.1093/biolinnean/blw014>, pp. 133–149.
- [12] **Gruenstaeudl, M**, L Nauheimer, and T Borsch. "Plastid genome structure and phylogenomics of Nymphaeales: Conserved gene order and new insights into relationships". In: *Plant Systematics and Evolution* 303 (2017). doi: <https://doi.org/10.1007/s00606-017-1436-5>, pp. 1251–1270.
- [11] N Korotkova, G Parolly, A Khachatryan, L Ghulikyan, H Sargsyan, J Akopian, T Borsch, and **Gruenstaeudl, M**. "Towards resolving the evolutionary history of Caucasian pears (*Pyrus*, Rosaceae) - Phylogenetic relationships, divergence times and leaf trait evolution". In: *Journal of Systematics and Evolution* 56 (2017). doi: <https://doi.org/10.1111/jse.12276>, pp. 35–47.
- [10] E Maharramova, I Huseynova, S Kolbaia, **Gruenstaeudl, M**, T Borsch, and LAH Muller. "Phylogeography and population genetics of the riparian relict tree *Pterocarya fraxinifolia* (Juglandaceae) in the South Caucasus". In: *Systematics and Biodiversity* 16 (2017). doi: <https://doi.org/10.1080/14772000.2017.1333540>, pp. 14–27.
- [9] BC Carstens, **Gruenstaeudl, M**, and NM Reid. "Community trees: Identifying codiversification in the Paramo dipteran community". In: *Evolution* 70 (2016). doi: <https://doi.org/10.1111/evo.12916>, pp. 1080–1093.
- [8] **Gruenstaeudl, M**. "WARACS: Wrappers to automate the reconstruction of ancestral character states". In: *Applications in Plant Sciences* 4 (2016). doi: <https://doi.org/10.3732/apps.1500120>, p. 1500120.

- [7] **Gruenstaeudl, M**, NM Reid, GL Wheeler, and BC Carstens. “Posterior predictive checks of coalescent models: P2C2M, an R package”. In: *Molecular Ecology Resources* 16 (2015). doi: <https://doi.org/10.1111/1755-0998.12435>, pp. 193–205.
- [6] **Gruenstaeudl, M**, CV Hawkes, A Santos-Guerra, and RK Jansen. “Preliminary investigations of correlated diversification between plants and their associated arbuscular mycorrhizal fungi in Macaronesia”. In: *Proceedings of the Amurga International Conferences on Island Biodiversity 2011*. Ed. by J Caujape-Castells, G Nieto-Feliner, and JM Fernandez-Palacios. Las Palmas, Spain: Fundacion Canaria Amurga Maspalomas, 2013, pp. 146–153. ISBN: 978-84-616-7394-0.
- [5] **Gruenstaeudl, M**, A Santos-Guerra, CV Hawkes, and RK Jansen. “Molecular survey of arbuscular mycorrhizal fungi associated with *Tolpis* on three Canarian islands (Asteraceae)”. In: *Vieraea* 41 (2013). doi: <http://dx.doi.org/10.31939/vieraea.2013.41.17>, pp. 233–252. ISSN: 0210-945X.
- [4] **Gruenstaeudl, M**, A Santos-Guerra, and RK Jansen. “Phylogenetic analyses of *Tolpis* Adans. (Asteraceae) reveal patterns of adaptive radiation, multiple colonization and interspecific hybridization”. In: *Cladistics* 29 (2013). doi: <https://doi.org/10.1111/cla.12005>, pp. 416–434.
- [3] V Funk, A Anderberg, B Baldwin, R Bayer, J Bonifacio, I Breitwieser, L Brouillet, R Carbajal, R Chan, A Coutinho, D Crawford, J Crisci, M Dillon, S Freire, M Galbany Casals, N Garcia-Jacas, B Gemeinholzer, **Gruenstaeudl, M**, HW Lack, and L Watson. “Compositae metatrees: the next generation”. In: *Systematics, Evolution and Biogeography of the Compositae*. Ed. by VA Funk, A Susanna, TF Stuessy, and R Bayer. Vienna, Austria: International Association For Plant Taxonomy (IAPT), 2009, pp. 747–777. ISBN: 978-39-501-7543-1.
- [2] **Gruenstaeudl, M**, E Urtubey, RK Jansen, R Samuel, MHJ Barfuss, and TF Stuessy. “Phylogeny of Barnadesioideae (Asteraceae) inferred from DNA sequence data and morphology”. In: *Molecular Phylogenetics and Evolution* 51 (2009). doi: <https://doi.org/10.1016/j.ympev.2009.01.023>, pp. 572–587.
- [1] TF Stuessy, E Urtubey, and **Gruenstaeudl, M**. “Barnadesioideae (Barnadesioideae)”. In: *Systematics, Evolution and Biogeography of the Compositae*. Ed. by V.A. Funk, A. Susanna, T.F. Stuessy, and R. Bayer. Vienna, Austria: IAPT, 2009, pp. 215–228. ISBN: 978-39-501-7543-1.

## — Scientific/Conference Presentations —

- [30] M Gruenstaeudl. *A snapshot of the quality and reproducibility of GenBank-archived organellar genomes in 2026*. Conference presentation, NISBRE 2026, Bethesda, Maryland, USA, 2026.
- [29] M Gruenstaeudl. *Assembly quality and sequencing coverage variation in organellar and cyanobacterial genomes on GenBank: Implications for phylogenetic research*. Invited seminar, Universität Wien, Vienna, Austria, 2026.
- [28] M Gruenstaeudl. *From cyanobacteria to chloroplasts: New ways to evaluate plastid and bacterial genome quality using bioinformatics*. Invited seminar, Pittsburg State University, Kansas, USA, 2026.
- [27] M Gruenstaeudl. *Moderator of first trainee presentation section*. Session moderator, K-INBRE Symposium 2026, Overland Park, USA, 2026.
- [26] M Gruenstaeudl. *Learning from every angle: 360-degree cameras transform blended genetics laboratory instruction*. Conference presentation, OLC Accelerate 2025, Orlando, Florida, USA, 2025.
- [25] M Gruenstaeudl. *Novel bioinformatic methods for assessing organelle and cyanobacterial genome quality*. Invited seminar, Universität Regensburg, Regensburg, Germany, 2025.

- [24] M Gruenstaeudl. *Developing bioinformatic tools for the data mining and quality control of archived organellar genomes: An ideal setting for training the next generation of bioinformaticians*. Conference presentation, Austrian Bioinformatics Workshop 2024, Graz, Austria, 2024.
- [23] M Gruenstaeudl. *Strategies to improve alignment accuracy in large-scale plant phylogenomic analyses*. Conference presentation, 20th International Botanical Congress, Madrid, Spain, 2024.
- [22] M Gruenstaeudl. *Using genomic data mining and bioinformatic software development to assess data quality of organellar genomes on GenBank*. Invited seminar, Wichita State University, Kansas, USA, 2024.
- [21] M Gruenstaeudl. *Assessing data quality among archived plastid genomes via novel software tools*. Conference presentation (online), Botanical Society of America Conference, Connecticut, USA, 2021.
- [20] M Gruenstaeudl. *Assessing sequence coverage and inverted repeat annotations among plastid genomes*. Workshop organizer (online), Botanical Society of America Conference, Connecticut, USA, 2021.
- [19] M Gruenstaeudl. *Exploring data quality among accumulated plastid genomes as a prerequisite for barcoding initiatives using whole plastomes*. Conference presentation (online), 19. Österreichische Botanik-Tagung, Krems, Austria, 2021.
- [18] M Gruenstaeudl. *On the need for quality control in the generation and analysis of complete plastid genomes*. Conference presentation (online), Deutsche Botanische Gesellschaft, Oldenburg, Germany, 2021.
- [17] M Gruenstaeudl. *The bioinformatic application of complete plastid genomes as plant DNA barcodes*. Conference presentation (online), Barcode of Life Initiative, Vienna, Austria, 2020.
- [16] M Gruenstaeudl. *Bioinformatic strategies to submit DNA sequences to public sequence repositories*. Workshop organizer, Gesellschaft für Biologische Systematik, Munich, Germany, 2019.
- [15] M Gruenstaeudl. *The monophyly of the water lilies remains uncertain*. Conference presentation, Gesellschaft für Biologische Systematik, Munich, Germany, 2019.
- [14] M Gruenstaeudl. *Bioinformatic tools for the submission of DNA sequences to public sequence databases*. Conference presentation, Deutsche Botanische Gesellschaft, Klagenfurt, Austria, 2018.
- [13] M Gruenstaeudl. *Computational strategies for the selection of data partitioning schemes in phylogenomics*. Conference presentation, Gesellschaft für Biologische Systematik, Vienna, Austria, 2018.
- [12] M Gruenstaeudl. *Phylogenomics done properly: How to partition your data*. Workshop organizer, Gesellschaft für Biologische Systematik, Vienna, Austria, 2018.
- [11] M Gruenstaeudl. *Plastid phylogenomics of Nymphaeaceae and development of bioinformatic tools*. Conference presentation, Genomics in Biodiversity Research, Berlin, Germany, 2017.
- [10] M Gruenstaeudl. *Statistical hybrid detection and its impact on inferring ancestral distribution areas*. Conference presentation, Dahlem Center of Plant Sciences, Berlin, Germany, 2016.
- [9] M Gruenstaeudl. *Model-based inferences in plant systematics: Examples from a diverse toolbox*. Invited seminar, Universität Leipzig, Leipzig, Germany, 2015.
- [8] M Gruenstaeudl. *Novel bioinformatic methods in model-based inference of plant phylogeography*. Conference presentation, VISCEA Ecology & Evolution Conference, Vienna, Austria, 2015.
- [7] M Gruenstaeudl. *Hybrid taxa can mislead phylogeographic analyses in Tolpis (Asteraceae)*. Conference presentation, Society for the Study of Evolution Conference, USA, 2014.

- [6] M Gruenstaeudl. *Analyzing the diversification of plants and AM fungi in the age of next-generation sequencing*. Invited seminar, University of Wageningen, The Netherlands, 2011.
- [5] M Gruenstaeudl. *Correlated diversification between arbuscular mycorrhizal fungi and their plant hosts*. Conference presentation, 9th International Mycological Conference, Edinburgh, UK, 2010.
- [4] M Gruenstaeudl. *Phylogenetic investigations of correlated diversification between vascular plants and their associated AM fungi*. Conference presentation, Flora Macaronesia International Symposium, Azores, Portugal, 2010.
- [3] M Gruenstaeudl. *Time and sequence of dispersal events in the Macaronesian island lineage Tolpis*. Conference presentation, Botanical Society of America Conference, Utah, USA, 2009.
- [2] M Gruenstaeudl. *Phylogenetic relationships in Tolpis (Asteraceae) and the utility of ETS sequence variation*. Conference presentation, Botany 2008 Conference, Vancouver, Canada, 2008.
- [1] M Gruenstaeudl. *Phylogenetic relationships in Barnadesioideae revisited*. Conference presentation, Botany & Plant Biology 2007 Conference, Chicago, USA, 2007.

## Honors & Awards for Research

|                                                                                              |                                       |      |
|----------------------------------------------------------------------------------------------|---------------------------------------|------|
| <b>Outstanding Research and Scholarly Activity Award</b><br>(university-wide)                | Fort Hays State University            | 2025 |
| <b>Outstanding Scholarship Award</b> (Werth College of Science, Technology, and Mathematics) | Fort Hays State University            | 2025 |
| <b>Scholarship of excellence</b>                                                             | Land Niederösterreich                 | 2012 |
| <b>Graduate student research award</b>                                                       | American Society of Plant Taxonomists | 2011 |
| <b>Graduate student research award</b>                                                       | Mycological Society of America        | 2010 |
| <b>Graduate student research award</b>                                                       | Botanical Society of America          | 2010 |

## Teaching

### List of University Courses Taught

#### Assistant Professor at Fort Hays State Univ.

- [66] **Genetics. Lectures.** BIOL325, 3 credit hours. Spring. 2026.
- [65] **Genetics. Laboratories.** BIOL325, 1 credit hour. Spring. 2026.
- [64] **Topics in Biology: Crafting Scientific Presentations.** Graduate course. BIOL607/G, 2 credit hours. Spring. 2026.
- [63] **Topics in Biology: Molecular Biology.** Graduate course. BIOL607/G, 4 credit hours. Spring. 2026.
- [62] **Botany. Lectures.** BIOL250, 3 credit hours. Spring. 2025.
- [61] **Botany. Laboratories.** BIOL250L, 1 credit hour. Spring. 2025.
- [60] **Genetics. Lectures.** BIOL325, 3 credit hours. Fall. 2025.
- [59] **Genetics. Laboratories.** BIOL325, 1 credit hour. Fall. 2025.
- [58] **Plant Anatomy. Lectures.** BIOL330, 3 credit hours. Fall. 2025.
- [57] **Plant Anatomy. Laboratories.** BIOL330L, 1 credit hour. Fall. 2025.

- [56] **Principles of Biology. Lectures.** BIOL180, 3 credit hours. Spring. 2025.
- [55] **Topics in Biology: Bioinformatics. Graduate course.** BIOL607/G, 3 credit hours. Fall. 2025.
- [54] **Genetics. Lectures.** BIOL325, 3 credit hours. Spring. 2024.
- [53] **Genetics. Lectures.** BIOL325, 3 credit hours. Fall. 2024.
- [52] **Genetics. Laboratories.** BIOL325, 1 credit hour. Spring. 2024.
- [51] **Genetics. Laboratories.** BIOL325, 1 credit hour. Fall. 2024.
- [50] **Principles of Biology. Lectures.** BIOL180, 3 credit hours. Fall. 2024.
- [49] **Botany. Lectures.** BIOL250, 3 credit hours. Fall. 2023.
- [48] **Botany. Laboratories.** BIOL250L, 1 credit hour. Fall. 2023.
- [47] **Genetics. Lectures.** BIOL325, 3 credit hours. Spring. 2023.
- [46] **Genetics. Laboratories.** BIOL325, 1 credit hour. Spring. 2023.
- [45] **Principles of Biology. Lectures.** BIOL180, 3 credit hours. Spring. 2023.
- [44] **Principles of Biology. Lectures.** BIOL180, 3 credit hours. Fall. 2023.
- [43] **Readings in Biology: The Impact of AI on Biology and Medicine. Graduate course.** BIOL482/BIOL882, 2 credit hours. Fall. 2023.
- [42] **Topics in Biology: Bioinformatics. Graduate course.** BIOL607/G, 3 credit hours. Fall. 2023.
- [41] **Topics in Biology: Molecular Biology. Graduate course.** BIOL607/G, 4 credit hours. Fall. 2023.

---

— **Lecturer at the Freie Universität Berlin** —

- [40] **Botany & Microbiology for Biochemists. Lectures.** Course no. 23700. Fall (online). 2021.
- [39] **Botany & Plant Physiology for Veterinary Sciences. Lectures.** Course no. 23760b-c. Fall (online). 2021.
- [38] **Genetics & Genomics. Lectures.** Course no. 23771a. Fall (online). 2021.
- [37] **Genetics & Genomics. Laboratories.** Course no. 23771b. Fall (online). 2021.
- [36] **Introduction to Botany & Biodiversity. Lectures.** Course no. 23106. Fall (online). 2021.
- [35] **Introduction to Botany & Biodiversity. Laboratories.** Course no. 23108a-e. Fall (online). 2021.
- [34] **Topics in Biology: Evolution. Graduate seminar.** Course no. 23653. Spring (online). 2021.
- [33] **Topics in Biology: Evolution. Graduate laboratories.** Course no. 23654a-b. Spring (online). 2021.
- [32] **Botany & Microbiology for Biochemists. Lectures.** Course no. 23700. Fall (online). 2020.
- [31] **Botany & Plant Physiology for Veterinary Sciences. Lectures.** Course no. 23760b-c. Fall (online). 2020.
- [30] **Genetics & Genomics. Lectures.** Course no. 23771a. Fall (online). 2020.
- [29] **Genetics & Genomics. Laboratories.** Course no. 23771b. Fall (online). 2020.
- [28] **Introduction to Botany & Biodiversity. Lectures.** Course no. 23106. Fall (online). 2020.
- [27] **Introduction to Botany & Biodiversity. Laboratories.** Course no. 23108a-e. Fall (online). 2020.
- [26] **Topics in Biology: Evolution. Graduate seminar.** Course no. 23653. Spring (online). 2020.
- [25] **Topics in Biology: Evolution. Graduate laboratories.** Course no. 23654a-b. Spring (online). 2020.
- [24] **Botany & Microbiology for Biochemists. Lectures.** Course no. 23700. Fall. 2019.
- [23] **Botany & Plant Physiology for Veterinary Sciences. Lectures.** Course no. 23760b-c. Fall. 2019.



- [22] **Genetics & Genomics. Lectures.** Course no. 23771a. Fall. 2019.
- [21] **Genetics & Genomics. Laboratories.** Course no. 23771b. Fall. 2019.
- [20] **Introduction to Botany & Biodiversity. Lectures.** Course no. 23106. Fall. 2019.
- [19] **Introduction to Botany & Biodiversity. Laboratories.** Course no. 23108a-e. Fall. 2019.
- [18] **Research in Bioinformatics. Laboratories.** Course no. 19400432. Spring. 2019.
- [17] **Topics in Biology: Evolution. Graduate seminar.** Course no. 23653. Spring. 2019.
- [16] **Topics in Biology: Evolution. Graduate laboratories.** Course no. 23654a-b. Spring. 2019.
- [15] **Botany & Microbiology for Biochemists. Lectures.** Course no. 23700. Fall. 2018.
- [14] **Botany & Plant Physiology for Veterinary Sciences. Lectures.** Course no. 23760b-c. Fall. 2018.
- [13] **Current Topics in Plant Systematics & Evolution. Graduate seminar.** Course no. 23815. Fall. 2018.
- [12] **Genetics & Genomics. Lectures.** Course no. 23771a. Fall. 2018.
- [11] **Genetics & Genomics. Laboratories.** Course no. 23771b. Fall. 2018.
- [10] **Introduction to Botany & Biodiversity. Lectures.** Course no. 23106. Fall. 2018.
- [9] **Introduction to Botany & Biodiversity. Laboratories.** Course no. 23108a-e. Fall. 2018.
- [8] **Botany & Microbiology for Biochemists. Lectures.** Course no. 23700. Fall. 2017.
- [7] **Botany & Plant Physiology for Veterinary Sciences. Lectures.** Course no. 23760b-c. Fall. 2017.
- [6] **Introduction to Botany & Biodiversity. Lectures.** Course no. 23106. Fall. 2017.
- [5] **Introduction to Botany & Biodiversity. Laboratories.** Course no. 23108a-e. Fall. 2017.
- [4] **Topics in Biology: Evolution. Graduate seminar.** Course no. 23653. Spring. 2017.
- [3] **Topics in Biology: Evolution. Graduate laboratories.** Course no. 23654a-b. Spring. 2017.
- [2] **Introduction to Botany & Biodiversity. Laboratories.** Course no. 23108a-e. Fall. 2016.
- [1] **Evolution & Biodiversity–Botany. Graduate lectures.** Course no. 23303a-e. Spring. 2015.

---

## — Student Supervision

### — Graduate Students

---

- [17] STUDENT: Louisa Acquah. ROLE: Primary thesis advisor. Fort Hays State University, 2023–2026.
- [16] STUDENT: David Esteban Bohorquez. ROLE: Primary thesis advisor. Fort Hays State University, 2025–ongoing.
- [15] STUDENT: Eka Giorgashvili. ROLE: Primary thesis advisor. Freie Universität Berlin, 2019–2020.
- [14] STUDENT: Thanina Hamitouche. ROLE: Primary thesis advisor. Fort Hays State University, 2025–ongoing.
- [13] STUDENT: Yannick Hartmaring. ROLE: Primary thesis advisor. Freie Universität Berlin, 2020–2021.
- [12] STUDENT: Nils Jenke. ROLE: Primary thesis advisor. Freie Universität Berlin, 2020–2021.
- [11] STUDENT: Buddha Thapa Magar. ROLE: Primary thesis advisor. Fort Hays State University, 2024–ongoing.
- [10] STUDENT: Jessica Röstel. ROLE: Primary thesis advisor. Freie Universität Berlin, 2019–2020.

### — Undergraduate Students

---

- [9] STUDENT: Elijah McCullough. ROLE: Capstone project advisor. Fort Hays State University, 2025–2026.



- [8] STUDENT: Lauren Perez. ROLE: Capstone project advisor. Fort Hays State University, 2026–ongoing.
- [7] STUDENT: Bryce Steffan. ROLE: Capstone project advisor. Fort Hays State University, 2025–2026.
- [6] STUDENT: Brandon Williams. ROLE: Capstone project advisor. Fort Hays State University, 2025.
- [5] STUDENT: Gregory M. Smith. ROLE: Capstone project advisor. Fort Hays State University, 2024.
- [4] STUDENT: Phongsavanh Mongkhonvilay. ROLE: Capstone project advisor. Fort Hays State University, 2023.
- [3] STUDENT: Tilman Mehl. ROLE: Thesis advisor. Freie Universität Berlin, 2020.
- [2] STUDENT: Yannick Hartmaring. ROLE: Thesis advisor. Freie Universität Berlin, 2019.
- [1] STUDENT: Nils Jenke. ROLE: Thesis advisor. Freie Universität Berlin, 2019.

## — Honors, Awards & Certificates for Teaching —

|                                                                                                                         |                                       |      |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------|
| <b>Certificate in Effective Teaching Practices</b>                                                                      | Assoc. of College and Univ. Educators | 2025 |
| 9-month (25-module) course in effective teaching practices on implementation of evidence-based instructional approaches |                                       |      |
| <b>Teaching grant—Experiential learning innovation</b>                                                                  | Fort Hays State University            | 2023 |
| <b>Teaching grant—Undergraduate research experience</b>                                                                 | Fort Hays State University            | 2023 |
| <b>Teaching grant—Industry 4.0</b>                                                                                      | Freie Universität Berlin              | 2018 |
| <b>Teaching award</b>                                                                                                   | Freie Universität Berlin              | 2017 |
| <b>Teaching assistant award</b>                                                                                         | University of Texas at Austin         | 2007 |

## Service

### — Leadership Training —

|                                                                                                 |                              |      |
|-------------------------------------------------------------------------------------------------|------------------------------|------|
| <b>LHH Executive Program for Leaders &amp; Managers</b>                                         | LHH   OTM Career Development | 2022 |
| 4-month training in communication, transition management, interviews, hearings, and assessments |                              |      |

### — Committee Service —

#### — University Committees —

|                                                           |                            |              |
|-----------------------------------------------------------|----------------------------|--------------|
| <b>Strategic Planning &amp; Improv. Committee</b> , Chair | Fort Hays State University | 2025–ongoing |
| <b>Faculty Senate</b> , Full Member                       | Fort Hays State University | 2025–ongoing |
| <b>Departm. Hiring Committee</b> , Chair                  | Fort Hays State University | 2025         |
| <b>Faculty Senate</b> , Alternate Member                  | Fort Hays State University | 2024–2025    |
| <b>Departm. Graduate Education Committee</b> , Member     | Fort Hays State University | 2024–ongoing |
| <b>Departm. Hiring Committee</b> , Member                 | Fort Hays State University | 2024         |
| <b>Departm. Scholarship Committee</b> , Member            | Fort Hays State University | 2024         |
| <b>Departm. Infrastructure Committee</b> , Member         | Freie Universität Berlin   | 2017–2018    |

---

## Peer-Review Service

---

### Funding Agencies

---

- Deutsche Forschungsgemeinschaft (DFG)

### Scientific Journals

---

- |                                            |                                   |
|--------------------------------------------|-----------------------------------|
| • Annals of Botany                         | • Molecular Ecology Resources     |
| • BMC Plant Biology                        | • Nordic Journal of Botany        |
| • Botanical Journal of the Linnean Society | • Plant Systematics and Evolution |
| • Frontiers in Plant Science               | • PLOS One                        |
| • GigaScience                              | • Systematic Botany               |
| • Mathematical Biosciences                 | • Taxon                           |
| • Mitochondrial DNA Part B                 | • Willdenowia                     |
| • Molecular Ecology                        |                                   |

---

## Scientific Memberships

---

- |                                                            |                                                                   |
|------------------------------------------------------------|-------------------------------------------------------------------|
| • International Society for Computational Biology (ISCB)   | • German Association of University Professors and Lecturers (DHV) |
| • International Association for Plant Taxonomy (IAPT)      | • German Botanical Society (DBG)                                  |
| • Austrian Scientists & Scholars in North America (ASCINA) | • Council on Undergraduate Research (CUR)                         |